

On-Policy Distillation with Best-of- N Teacher Rollout Selection

Ke Zhang*
Johns Hopkins University

Yunjie Tian
TikTok

Dongdi Zhao
TikTok

Yijiang Li
University of California, San Diego

Yuanye Liu
Fudan University

Vishal M. Patel
Johns Hopkins University

Di Fu†
TikTok

Abstract

On-policy distillation (OPD), which supervises a student on its own sampled trajectories, has emerged as a data-efficient post-training method for improving reasoning while avoiding the reward dependence of reinforcement learning and the catastrophic forgetting often observed in standard supervised fine-tuning. However, standard OPD typically computes teacher supervision under noisy student-generated contexts and often relies on a single stochastic teacher rollout per prompt. As a result, the supervision signal can be high-variance: the sampled teacher trajectory can be incorrect, uninformative, or poorly matched to the student’s current reasoning behavior. To address this limitation, we propose **BRTS**, a **B**est-of- N **R**ollout **T**eacher **S**election framework for on-policy distillation. BRTS augments standard student-context OPD with a teacher-context supervision branch constructed from the curated teacher trajectory. Rather than distilling from the first sampled teacher rollout, BRTS samples a small pool of teacher trajectories and selects the auxiliary trajectory using a simple priority rule: correctness first, student alignment second. When multiple correct teacher trajectories are available, BRTS chooses the one most aligned with the student’s current behavior; when unconditioned teacher samples fail on harder prompts, it invokes a ground-truth-conditioned recovery step to elicit a natural derivation. The selected trajectory is then used to provide reliable teacher-context supervision inside the OPD loop, augmented with an auxiliary loss on the teacher trajectory. Experiments on AIME 2024, AIME 2025, and AMC 2023 show that BRTS improves over standard OPD on challenging reasoning benchmarks, with the largest gains on harder datasets. Our code is available at <https://github.com/BWGZK-keke/BRTS>.

1 Introduction

On-policy distillation (OPD) has rapidly become a standard tool in large language model (LLM) post-training [1, 13, 19, 21, 24, 29, 44, 47, 50, 51]. The recipe is appealingly simple: have the student generate rollouts from its own policy, and use the teacher’s per-token log-probabilities on those rollouts as a dense supervision signal. Compared with supervised fine-tuning (SFT) on teacher-generated text or classical sequence-level distillation [12, 17, 20, 23, 34, 42], OPD is less susceptible to exposure bias because supervision is defined on exactly the states the student visits at inference time [3, 33, 45]. Industry post-training pipelines [11, 44, 47] have adopted OPD as a competitive complement to SFT and outcome-reward reinforcement learning [4, 5, 15, 35, 37, 52],

*Work done during an internship at TikTok.

†Corresponding author.

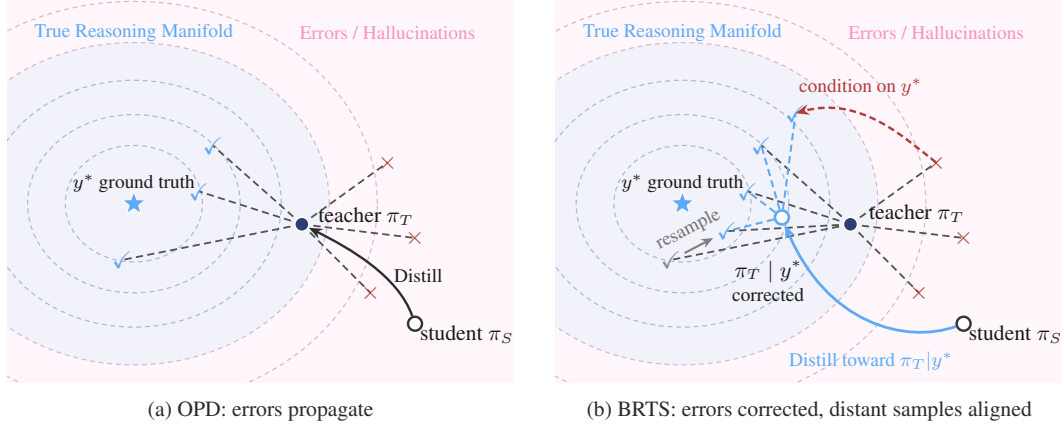


Figure 1: Conceptual comparison of OPD and BRTS. (a) Classical OPD may propagate unreliable teacher signals when teacher trajectories are incorrect or poorly matched to the student. (b) BRTS constructs correctness- and alignment-aware teacher-context supervision by selecting or recovering a reliable teacher trajectory before distillation.

with reported comparable gains at a fraction of the RL compute cost [29]. Despite this appeal, recent work has begun to unpack when OPD works and why [10, 22, 26]. Two conditions appear to govern success: the student and teacher should share compatible reasoning patterns, and the teacher be able to produce correct, naturally derived solutions beyond the student’s current exploration range, thereby transferring new capabilities on difficult tasks. This observation is also consistent with recent findings that small models can struggle to imitate strong reasoners when the teacher’s reasoning style is poorly matched to the student [9, 14, 27].

Standard OPD reduces exposure bias by evaluating teacher feedback on student-generated trajectories, but this also means that supervision is computed on prefixes produced by an imperfect student [1, 3, 29, 33]. On difficult prompts, these prefixes can drift into noisy reasoning states, where local teacher feedback is less informative, and the student may never observe a complete correct solution path [10, 26, 27]. Practical OPD-style pipelines therefore often use teacher-side signals, such as reference rollouts, solvability estimates, metadata, or filtering rules, to identify prompts or trajectories that provide useful supervision [26, 29, 44, 49]. However, the teacher is also stochastic: for a fixed prompt, its sampled trajectories can differ substantially in correctness, reasoning style, and proximity to the student, especially on hard prompts [28, 39, 43, 53]. Thus, a single teacher sample can provide a high-variance estimate of the teacher’s competence and alignment for a given prompt. When this sample is incorrect or poorly matched to the student, it can produce noisy guidance (Figure 1 (a)).

To address this limitation, we introduce **BRTS**, a **B**est-of- N **R**ollout **T**eacher **S**election framework for on-policy distillation. As illustrated in Figure 1(b), BRTS first identifies a reliable teacher trajectory and then distills it as an additional teacher-context supervision signal, rather than learning from an arbitrary teacher rollout. This complements standard student-context distillation: while OPD corrects the states visited by the student, BRTS exposes the student to complete and reliable teacher reasoning paths. The teacher-context branch is both correctness-aware and alignment-aware. Correctness prevents erroneous teacher samples from being amplified, while alignment keeps the selected trajectory close to the student’s current reasoning distribution. For prompts where ordinary teacher sampling fails, BRTS uses a ground-truth-guided recovery step to obtain a teacher derivation, keeping the auxiliary branch active on hard examples where reliable supervision is most needed. In this way, BRTS turns stochastic teacher rollouts into structured supervision inside the OPD loop. Experiments on three public datasets show that BRTS improves over standard OPD, with larger gains on harder prompts where reliable teacher trajectories are sparse.

Our contributions are summarized as follows:

- We identify a high-variance supervision issue in OPD: standard student-context supervision can be limited by noisy student prefixes and stochastic teacher rollouts, making the teacher signal unreliable on hard prompts.

- We introduce *BRTS*, a Best-of- N rollout teacher selection framework that provides a lightweight mechanism for selecting and distilling reliable teacher trajectories. With an answer-hint fallback for failed unconditioned samples, BRTS distills the selected trajectory through a ground truth-guided teacher-context objective.
- BRTS significantly improves over standard OPD on the hard benchmarks. We also provide diagnostic analyses showing how teacher candidate quality, recovery coverage, and teacher backbone choice affect the resulting OPD performance.

2 Related Work

From off-policy to on-policy distillation. Knowledge distillation transfers behavior and capabilities from a stronger teacher to a smaller or weaker student [12, 17, 20, 23, 34, 42]. For autoregressive language models, standard off-policy distillation on teacher-generated text introduces a distribution mismatch: the student is trained under teacher-induced contexts but must condition on its own prefixes at inference time [3, 33, 45]. Optimizing supervised objectives on fixed teacher outputs may further over-constrain the student to trajectories outside its reachable distribution, increasing learning difficulty [46, 54] and causing catastrophic forgetting [30, 36]. On-policy distillation (OPD) alleviates this by letting the student generate trajectories and using the teacher to supervise the states the student actually visits. MiniLLM [13] and GKD [1] formulate this with reverse-KL and related divergences, while recent systems such as Qwen3 [47], MiMo [44], and GLM-5 [11] combine distillation with supervised fine-tuning and outcome-reward RL [15, 35, 52]. Recent variants extend OPD to black-box, contextual, entropy-aware, and reward-extrapolated settings [21, 29, 49–51].

Understanding and improving on-policy distillation. Recent work has begun to analyze when OPD succeeds or fails. [26] shows that successful OPD requires compatible student–teacher thinking patterns, genuinely new teacher knowledge, and increasing overlap between their high-probability token sets. Other studies identify failures caused by noisy student prefixes, entropy mismatch, unstable targets, or weak alignment between the student’s reasoning state and teacher feedback [10, 19, 21, 22]. These findings suggest that OPD depends not only on the divergence, but also on the *context* where teacher supervision is applied. This further relates to the token-level loss direction: classical KD emphasizes teacher-confident tokens [17], whereas many OPD implementations supervise tokens sampled from, or highly weighted by, the student policy [29, 44].

Rollout selection, privileged hints, and data filtering. Our method is related to best-of- N sampling, rejection sampling, and self-training methods that generate multiple candidates and retain high-quality outputs for inference or training [6, 8, 28, 31, 39, 41, 43, 53]. Unlike these outer-loop filtering approaches, BRTS performs selection inside the OPD loop: from a small set of teacher rollouts, it chooses an auxiliary trajectory by prioritizing correctness and then proximity to the current student. The selected rollout is thus used not as offline training data, but as a teacher-context signal tailored to the current OPD update. BRTS also connects to methods that use privileged information, such as ground-truth answers, demonstrations, or feedback, to strengthen supervision [7, 18, 32, 38, 40, 48, 55].

3 Method

3.1 Preliminaries

Let x be an input prompt and y^* be its corresponding ground-truth answer. We consider a student policy π_S and a teacher policy π_T , each defining an autoregressive next-token distribution over a vocabulary $\mathcal{V}$. For a trajectory $y = (y_1, \dots, y_T)$, the probability under a policy π factorizes as

$$\pi(y \mid x) = \prod_{t=1}^T \pi(y_t \mid x, y_{<t}), \quad (1)$$

where $y_{<t}$ denotes the prefix before token t . Let $\hat{y}^S \sim \pi_S(\cdot \mid x)$ and $y^T \sim \pi_T(\cdot \mid x)$ denote student and teacher rollouts, respectively. On-policy distillation trains the student on states induced by its own generation: for rollout $\hat{y}^S$, the teacher provides token-level supervision on prefixes $(x, \hat{y}_{<t}^S)$. A standard formulation minimizes the sequence-level reverse-KL objective [1, 13, 26]:

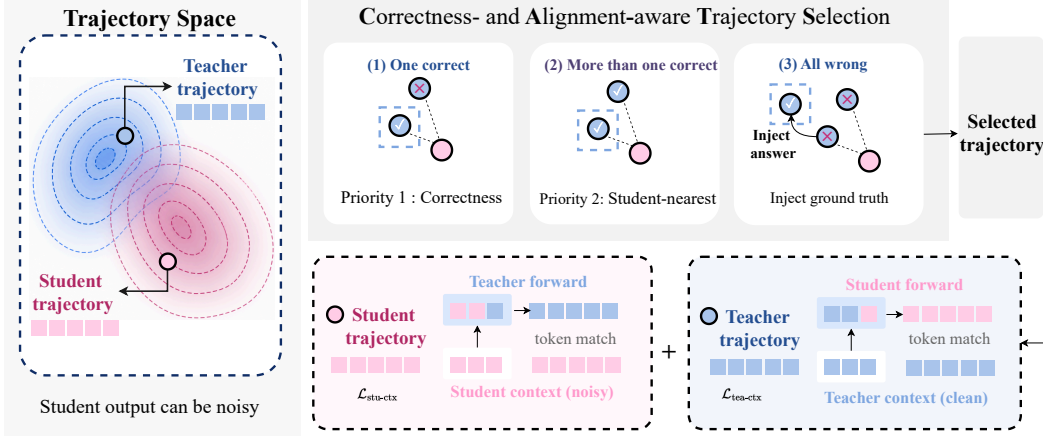


Figure 2: Overview of BRTS. The left panel shows teacher and student trajectories in the trajectory space. The upper-right panel illustrates the selection rule: choose a correct teacher rollout when available, select the student-nearest one when multiple correct rollouts exist, and inject the ground-truth answer when all sampled rollouts are wrong. The lower panels describe the loss computation, where token matching is performed under noisy student context and clean teacher context, respectively.

$$\mathcal{L}_{\text{OPD}}(S) = \mathbb{E}_{x, \hat{y}^S \sim \pi_S} \left[\sum_{t=1}^T D_{\text{KL}}(\pi_S(\cdot | x, \hat{y}_{<t}^S) \| \pi_T(\cdot | x, \hat{y}_{<t}^S)) \right]. \quad (2)$$

In OPD implementations, the per-step KL is often approximated with a sampled or top- K token set, typically chosen from the student distribution at the current student prefix [29, 44]. This corrects the student on the states it actually visits. However, the student context can be noisy. BRTS addresses this by adding a correctness- and alignment-aware teacher-context branch. We describe BRTS in three parts: teacher trajectory curation (Sec. 3.2), teacher-context supervision (Sec. 3.3), and the top- K candidate direction used in each branch (Sec. 3.4). Figure 2 shows our framework and Algorithm 1 summarizes the training step.

3.2 Correctness- and Alignment-aware Trajectory Curation

In the first tier, we draw N unconditioned teacher samples $\{y^{T,i}\}_{i=1}^N \sim \pi_T(\cdot | x)$ for each prompt. Each rollout is evaluated against the ground-truth answer y^* by extracting its final answer. If at least one rollout is correct, *i.e.*, $\text{answer}(y^{T,i}) = y^*$, BRTS selects among the correct rollouts the trajectory with the highest token-level overlap with the student’s top- K candidate set. This alignment criterion favors correct trajectories that remain close to the student’s high-probability region and are therefore more suitable for distillation. If none of the unconditioned teacher rollouts is correct, BRTS invokes a ground-truth-guided recovery step. We construct a modified prompt x^{gt} that provides the correct answer y^* as a silent validation signal while instructing the teacher to produce a natural derivation. A guided teacher rollout $y^{T,\text{gt}} \sim \pi_T(\cdot | x^{\text{gt}})$ is then sampled and retained only if its extracted answer is correct. This step targets hard prompts where ordinary teacher sampling fails, allowing the teacher-context branch to remain active on examples where reliable supervision is most needed. If neither unconditioned sampling nor guided recovery yields a correct trajectory, BRTS falls back to the Tier-1 rollout with the highest student overlap. Although this fallback trajectory may be incorrect, selecting the closest available rollout avoids introducing an arbitrary teacher trajectory far from the student’s distribution.

3.3 Teacher-Context Supervision

We retain Eq. (2) on student-generated trajectories and augment it with a teacher-context distillation loss defined on the curated teacher trajectory y' from Sec. 3.2. Both branches compare teacher and student distributions under matched conditioning contexts. In the student-context branch, both distributions are conditioned on the student prefix $\hat{y}_{<t}^S$. In the teacher-context branch, both distributions

Algorithm 1 BRTS training step

Require: Prompt x , ground-truth answer y^* , teacher policy π_T , student policy π_S , number of Tier-1 rollouts N , auxiliary weight λ

- 1: Sample a student rollout $\hat{y}^S \sim \pi_S(\cdot | x)$
- 2: Draw N unconditioned teacher rollouts $\{y^{T,i}\}_{i=1}^N \sim \pi_T(\cdot | x)$
- 3: Grade each teacher rollout by checking whether $\text{answer}(y^{T,i}) = y^*$
- 4: **if** at least one Tier-1 rollout is correct **then**
- 5: Select y' as the correct rollout with the highest top- K overlap with the student trajectory.
- 6: **else**
- 7: Construct the ground-truth-guided prompt x^{gt}
- 8: Sample one guided teacher rollout $y^{T,\text{gt}} \sim \pi_T(\cdot | x^{\text{gt}})$
- 9: **if** $\text{answer}(y^{T,\text{gt}}) = y^*$ **then**
- 10: Select $y' \leftarrow y^{T,\text{gt}}$
- 11: **else**
- 12: Select y' as the Tier-1 rollout with the highest student top- K overlap
- 13: **end if**
- 14: **end if**
- 15: Compute the student-context loss $\mathcal{L}_{\text{stu-ctx}}$ on $\hat{y}^S$ using student top- K candidates
- 16: Compute the teacher-context loss $\mathcal{L}_{\text{tea-ctx}}$ on the selected trajectory y' using teacher top- K candidates
- 17: $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{stu-ctx}} + \lambda \mathcal{L}_{\text{tea-ctx}}$
- 18: Update the student parameters by taking a gradient step on $\mathcal{L}_{\text{total}}$

are conditioned on the selected teacher prefix $y'_{<t}$. The resulting objectives are

$$\mathcal{L}_{\text{stu-ctx}} = \mathbb{E} \left[\sum_t D_{\text{KL}}(\pi_S(\cdot | x, \hat{y}_{<t}^S) \| \pi_T(\cdot | x, \hat{y}_{<t}^S)) \right], \quad (3)$$

$$\mathcal{L}_{\text{tea-ctx}} = \mathbb{E} \left[\sum_t D_{\text{KL}}(\pi_T(\cdot | x, y'_{<t}) \| \pi_S(\cdot | x, y'_{<t})) \right], \quad (4)$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{stu-ctx}} + \lambda \mathcal{L}_{\text{tea-ctx}}. \quad (5)$$

The two branches play complementary roles. The student-context loss preserves the on-policy nature of OPD by correcting the states visited by the student. The teacher-context loss exposes the student to a coherent teacher reasoning path selected for both correctness and compatibility with the student’s current distribution. Thus, BRTS does not replace OPD; it augments OPD with a structured teacher-context signal that is less sensitive to stochastic teacher failures. The coefficient λ controls the strength of the teacher-context branch. Since this branch is evaluated on selected teacher trajectories that are usually more coherent than noisy student rollouts, its raw contribution can be small with the naive choice $\lambda = 1$. Empirically, we find that $\lambda = 10$ gives the auxiliary branch a meaningful scale while remaining stable, and use this value across all experiments.

3.4 Top- K Direction: Student- vs. Teacher-Confident

In the sampled-token setting, both objectives are implemented as top- K aggregation losses, but they differ in how the candidate token set is defined. For the student-context branch, the candidate set consists of the student’s top- K tokens under the student prefix. This branch therefore supervises the tokens that the student considers plausible, providing targeted correction on student-visited states. For the teacher-context branch, the candidate set consists of the teacher’s top- K tokens under the selected teacher prefix. Unlike the student-context branch, this set reflects the teacher’s local distribution along a high-quality trajectory. Thus, the auxiliary branch introduces teacher-preferred tokens that may lie outside the student’s current top choices, while still keeping the supervision concentrated on a compact top- K candidate set.

Table 1: **Effect of the teacher-context supervision.** The baseline uses two student rollouts. Our variants replace one student rollout with one auxiliary teacher trajectory selected from N teacher candidates. We report mean, best, and majority accuracy on AIME24, AIME25, and AMC23.

Rollout Setup			AIME24			AIME25			AMC23		
$\mathcal{L}_{\text{tea-ctx}}$	Student	Teacher / Cand.	Mean	Best	Majority	Mean	Best	Majority	Mean	Best	Majority
×	2	0 / 0	0.3917	0.5915	0.4146	0.2667	0.3503	0.2792	0.6777	0.7959	0.6952
✓	1	1 / 1	0.3750	0.5805	0.4124	0.2583	0.3823	0.2799	0.6747	0.8047	0.6978
✓	1	1 / 2	0.3750	0.5855	0.3953	0.2667	0.4056	0.2748	0.6717	0.8104	0.6917
✓	1	1 / 4	0.4000	0.5991	0.4306	0.2917	0.3700	0.2961	0.6837	0.8004	0.6948

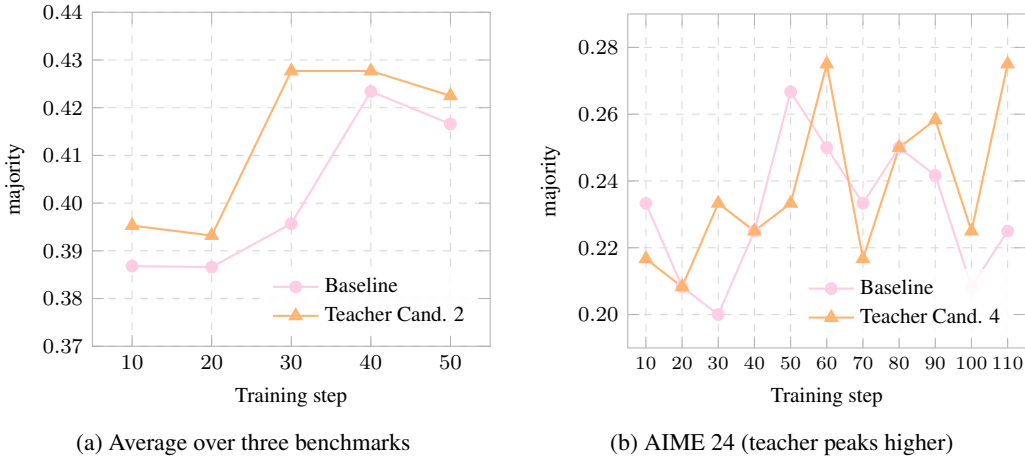


Figure 3: Majority-vote accuracy across training steps. The student-only baseline uses two student rollouts, while BRTS replaces one student rollout with an auxiliary teacher trajectory selected from a small candidate pool. Left: average over AIME24, AIME25, and AMC23. Right: AIME24 results. BRTS achieves a higher early-training peak than the student-only baseline.

4 Experiments

4.1 Setup

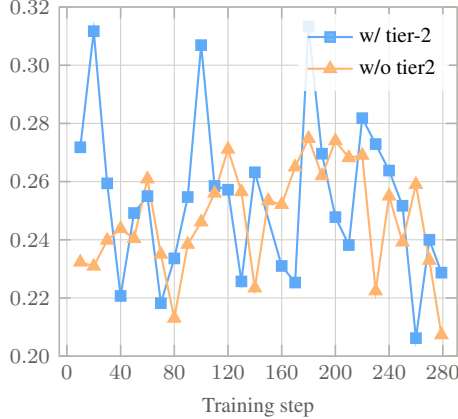
We follow the standard OPD training recipe used in recent work [26, 29]. Our main experiments use JustRL-1.5B [16] as the teacher and a same-scale student backbone DeepSeek-1.5B [15], trained on instructions sampled from DAPO-Math-17K [52]. To examine whether the selector depends on a specific teacher family, we further conduct a teacher-swap experiment with DeepSeek-R1-Distill-Qwen-7B [15] as the teacher. We run all main training experiments on $8 \times \text{B200}$ GPUs. We evaluate mathematical reasoning on AIME 2024 [25], AIME 2025 [2], and AMC 2023 [25]. AIME provides challenging short-answer problems requiring multi-step reasoning and exact numerical answers, while AMC 2023 offers relatively easier multiple-choice problems covering broader contest-math topics. We include both AIME 2024 and 2025 to test robustness across contest years, with AIME 2025 serving as a more difficult evaluation set. By default, we sample $k = 4$ solutions per problem with temperature 0.7 and top- $p = 0.95$, and report the average accuracy across k sampled solutions, the best-of- k accuracy, and the majority-vote accuracy. More details are provided in Appendix A.

4.2 Experiment Results

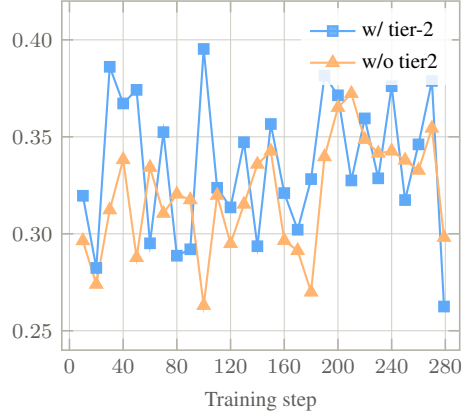
Effect of the teacher-context loss. We first evaluate whether teacher rollouts provide additional supervision beyond the standard student-only rollout. Table 1 compares different rollout configurations on AIME24, AIME25, and AMC23. Overall, teacher rollouts improve performance on the more challenging AIME benchmarks. Compared with the student-only setting, replacing one student rollout with a selected teacher rollout and enlarging the teacher candidate pool generally yields more informative supervision. For example, using four teacher candidates achieves the best AIME24 results, reaching 0.400 average accuracy, 0.599 best accuracy, and 0.4306 majority-vote accuracy.

Table 2: **Effect of Tier-2 ground-truth-guided selection.** Tier 1 uses unconditioned teacher samples, while Tier 2 adds ground-truth-guided re-sampling for harder prompts not solved by Tier 1. We report mean, best, and majority-vote accuracy on AIME25 and AMC23.

$\mathcal{L}_{\text{tea-ctx}}$	Tier 2	Rollout Setup				AIME25			AMC23		
		Student	Teacher / Cand.	Tier 1	Tier 2	Mean	Best	Majority	Mean	Best	Majority
×	×	2	0 / 0	0	0	0.2667	0.3503	0.2792	0.6777	0.7959	0.6952
✓	×	1	1 / 2	2	0	0.2667	0.4056	0.2748	0.6717	0.8104	0.6917
✓	×	1	1 / 4	4	0	0.2917	0.3700	0.2961	0.6837	0.8004	0.6948
✓	✓	1	1 / 2+1	2	1	0.3000	0.4309	0.3133	0.6747	0.8066	0.6997



(a) AIME 25



(b) AIME 24

Figure 4: Majority accuracy on AIME25 and AIME24 across training steps, comparing Tier-1 selection with two teacher candidates to the same setting augmented with Tier-2 recovery. Tier-2 substantially improves performance on the harder AIME25 benchmark, suggesting that teacher-guided correction is most beneficial for difficult tasks.

This suggests that sampling multiple teacher trajectories increases the likelihood of selecting a useful reasoning path. Figure 3 further shows that replacing one student rollout with a teacher rollout leads to a substantially higher early-stage performance peak. This indicates that a selected teacher rollout can provide more informative supervision than an additional student rollout. On AMC23, the gains are more modest, likely because the student-only baseline is already strong. Nevertheless, using four teacher rollout candidates still improves average accuracy, suggesting that a larger teacher candidate pool can provide useful supervision even on easier benchmarks.

Effect of Tier-2 ground-truth-guided selection. We further analyze Tier-2 selection, where a ground-truth-guided hint is injected into the teacher prompt when all initial teacher samples fail. Figures 4a and 4b compare training with and without Tier-2 on AIME25 and AIME24. On the harder AIME25 benchmark, Tier-2 consistently improves majority-vote accuracy at early training stages. It also achieves the best results in Table 2, reaching 0.3000 average accuracy, 0.4309 best accuracy, and 0.3133 majority-vote accuracy. This suggests that hint-guided teacher correction is particularly useful when unconditioned teacher rollouts fail to provide reliable supervision.

Computational Cost. Increasing the teacher candidate pool mainly affects the selection stage: each additional candidate adds about 59 seconds per step in our setup, as detailed in Appendix C, which remains acceptable. Importantly, it does not increase the number of trajectories used in the loss. In the two-rollout setting, for example, BRTS replaces one student rollout with one selected teacher rollout. Since teacher solutions are often more confident and direct, they tend to produce shorter sequences. BRTS also improves the compute-performance trade-off by achieving stronger performance in early training. As shown in Figures 3 and 4, replacing one student rollout with a selected teacher rollout leads to a higher early-training peak, and Tier-2 further strengthens this effect on harder tasks. Since longer distillation may suffer from catastrophic forgetting, reaching better performance earlier can enable shorter training and reduce total computation. Thus, BRTS trades a

Table 3: Effect of applying a prompt perturbation during teacher rollout, on AMC 2023.

Metric	Method	Step 10	Step 20	Step 30	Step 40	Step 50
AMC mean	w/o perturb	0.6175	0.6416	0.6416	0.6446	0.6325
	w/ perturb	0.6566	0.6566	0.6476	0.6265	0.6476
AMC majority	w/o perturb	0.6339	0.6663	0.6570	0.6721	0.6531
	w/ perturb	0.6759	0.6764	0.6615	0.6360	0.6698

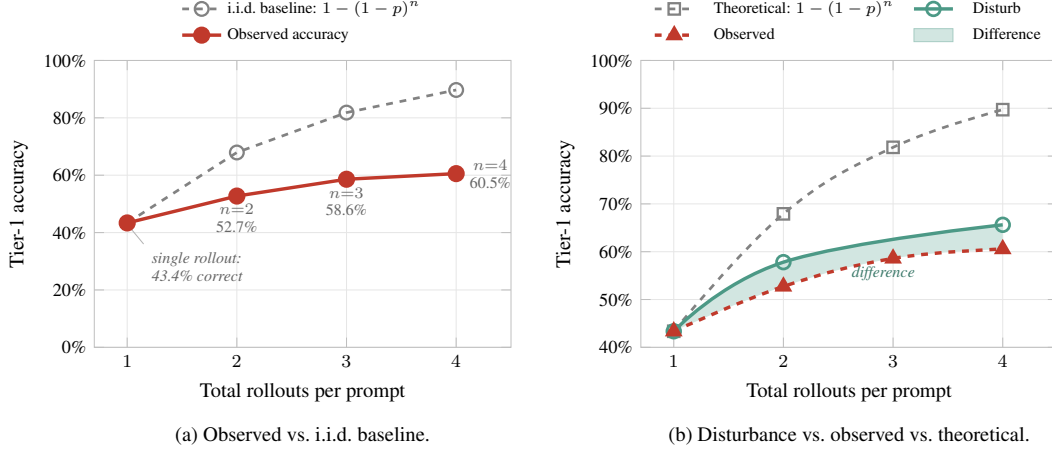


Figure 5: Tier-1 accuracy as a function of total rollouts per prompt. Panel (a) compares observed accuracy against the i.i.d. baseline $1 - (1 - p)^n$. Panel (b) shows the gap between rollouts with one disturbance, the observed accuracy, and the theoretical i.i.d. curve.

modest teacher-sampling overhead for higher-quality supervision, faster convergence, and a higher potential performance upper bound.

Prompt Perturbation. We further examine the sensitivity of BRTS to teacher prompt wording by applying a small prompt perturbation to one of the two teacher candidates. Detailed prompts are provided in Appendix B. As shown in Table 3, this perturbation does not degrade early performance on AMC23; instead, it raises early step best average accuracy from 0.6416 to 0.6566 and majority-vote accuracy from 0.6663 to 0.6764. Figure 5 helps explain this effect. Without added diversity, the Tier-1 catch rate grows much more slowly than the ideal i.i.d. prediction $1 - (1 - p)^n$, suggesting strong correlation among teacher samples. The perturbed rollout shifts the observed curve upward, indicating that mild prompt variation partially decorrelates teacher trajectories and increases the chance of finding a correct one. Overall, BRTS is not brittle to a single teacher prompt. Since selection is correctness-filtered, mild prompt perturbation can safely improve rollout diversity while preserving reliable supervision, especially in early training.

Backbones. To evaluate whether BRTS transfers across teacher backbones, we use DeepSeek-R1-Distill-Qwen-7B as the teacher and DeepSeek-1.5B as the student, focusing on early training steps. As shown in Table 4, BRTS improves most metrics over the student-only baseline at this early stage. On AIME24, it increases average accuracy from 0.3167 to 0.3667 and majority-vote accuracy from 0.3331 to 0.3952. On AIME25, all three metrics improve, with average, best, and majority-vote accuracy rising from 0.2167, 0.2770, and 0.2277 to 0.2417, 0.3294, and 0.2528, respectively. The gains also transfer to AMC23, where average, best, and majority-vote accuracy improve from 0.6265, 0.7410, and 0.6453 to 0.6446, 0.7836, and 0.6676. These results suggest that BRTS is not tied to a single teacher family: even after changing the teacher backbone, correctness-selected teacher rollouts continue to provide stronger supervision than the on-policy-only baseline.

Teacher Candidate Composition. We decompose the selected teacher trajectories into Tier-1 and Tier-2 sources. Tier-1 corresponds to prompts solved by unconditioned teacher sampling, Tier-2 corresponds to prompts rescued by ground-truth-guided re-sampling, and the remaining cases use

Table 4: Backbone ablation with DeepSeek-R1-Distill-Qwen-7B as the teacher at early step.

Rollout Setup		AIME24			AIME25			AMC23		
Student	Teacher / Cand.	Mean	Best	Maj	Mean	Best	Maj	Mean	Best	Maj
2	0 / 0	0.3167	0.4499	0.3331	0.2167	0.2770	0.2277	0.6265	0.7410	0.6453
1	1 / 2	0.3667	0.4485	0.3952	0.2417	0.3294	0.2528	0.6446	0.7836	0.6676

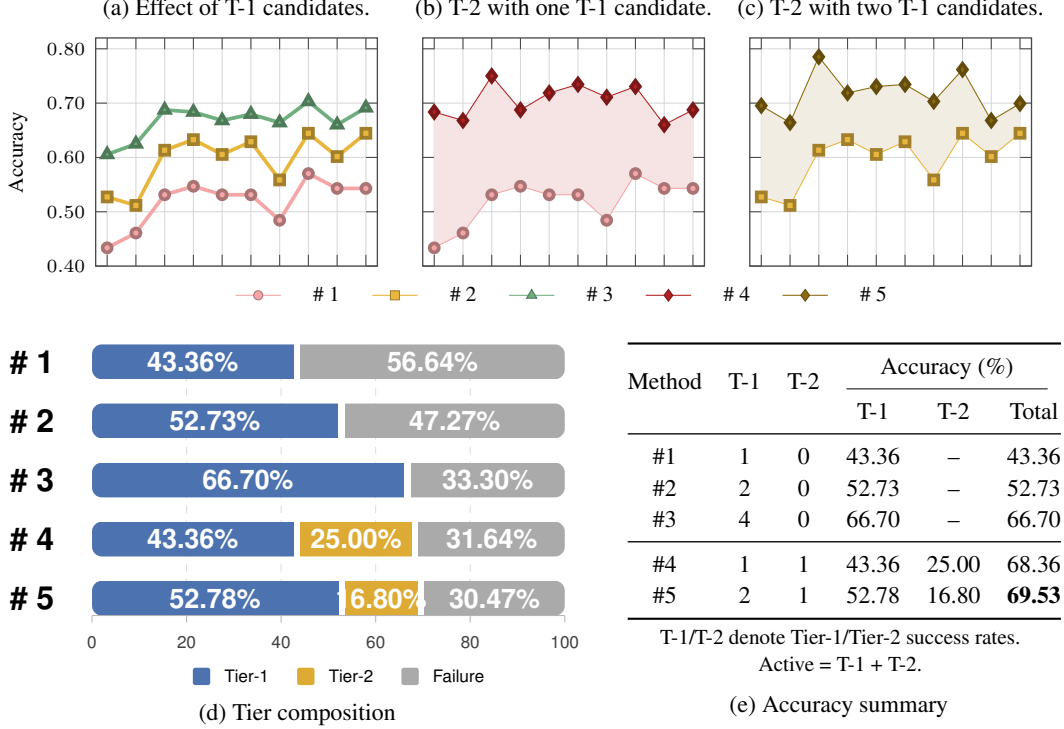


Figure 6: *Top*: Accuracy across training steps. Adding more Tier-1 candidates improves unconditioned teacher coverage (a), while Tier-2 further increases the catch rate by recovering prompts missed by Tier-1 (b,c). *Bottom*: Tier composition example, where Tier-1 is unconditioned teacher success, Tier-2 is ground-truth-guided recovery, and grey denotes fallback cases.

the fallback trajectory. As shown in Figure 6, using more Tier-1 candidates improves unconditioned coverage: the Tier-1 accuracy rate increases from 52.73% with two candidates to 66.70% with four candidates. However, this improvement requires additional teacher sampling budget on every prompt. Tier-2 provides a more targeted alternative because it is only applied when the Tier-1 candidates fail. With one Tier-1 candidate, Tier-2 recovers an additional 25.00% of prompts, increasing the total accuracy from 43.36% to 68.36%. With two Tier-1 candidates, Tier-2 recovers another 16.80% of prompts, raising the accuracy from 52.78% to 69.53%. Results show that adding Tier-1 candidates increases the chance of sampling a correct trajectory, while Tier-2 recovers prompts that unconditioned sampling fails to solve.

5 Conclusion

We presented BRTS, a lightweight and effective extension to on-policy distillation that improves how teacher trajectories are selected and used for supervision. Rather than distilling from a random teacher rollout, BRTS samples multiple candidate trajectories, filters for correctness, and selects a rollout that is better aligned with the student. The selected trajectory is then used by a teacher-context auxiliary branch with teacher-top- K supervision, allowing the student to learn from higher-quality teacher states. Experiments on AIME 2024, AIME 2025, and AMC 2023 show that BRTS provides consistent gains over the OPD baseline.

References

- [1] Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos Garea, Matthieu Geist, and Olivier Bachem. On-policy distillation of language models: Learning from self-generated mistakes. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL <https://openreview.net/forum?id=3zKtaqxLhW>.
- [2] Mislav Balunović, Jasper Dekoninck, Ivo Petrov, Nikola Jovanović, and Martin Vechev. MathArena: Evaluating LLMs on uncontaminated math competitions. *arXiv preprint arXiv:2505.23281*, 2025.
- [3] Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
- [4] Howard Chen, Noam Razin, Karthik Narasimhan, and Danqi Chen. Retaining by doing: The role of on-policy data in mitigating forgetting. *arXiv preprint arXiv:2510.18874*, 2025.
- [5] Tianzhe Chu, Yuexiang Zhai, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V. Le, Sergey Levine, and Yi Ma. SFT memorizes, RL generalizes: A comparative study of foundation model post-training. *arXiv preprint arXiv:2501.17161*, 2025.
- [6] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- [7] Ken Ding. HDPO: Hybrid distillation policy optimization via privileged self-distillation. *arXiv preprint arXiv:2603.23871*, 2026.
- [8] Hanze Dong, Wei Xiong, Deepanshu Goyal, Yihan Zhang, Winnie Chow, Rui Pan, Shizhe Diao, Jipeng Zhang, KaShun Shum, and Tong Zhang. RAFT: Reward ranked finetuning for generative foundation model alignment. In *Transactions on Machine Learning Research*, 2023.
- [9] Yao Fu, Hao Peng, Litu Ou, Ashish Sabharwal, and Tushar Khot. Specializing smaller language models towards multi-step reasoning. *arXiv preprint arXiv:2301.12726*, 2023.
- [10] Yuqian Fu, Haohuan Huang, Kaiwen Jiang, Yuanheng Zhu, and Dongbin Zhao. Revisiting on-policy distillation: Empirical failure modes and simple fixes. *arXiv preprint arXiv:2603.25562*, 2026.
- [11] GLM-5 Team, Aohan Zeng, Xin Lv, Zhenyu Hou, et al. GLM-5: From vibe coding to agentic engineering. *arXiv preprint arXiv:2602.15763*, 2026.
- [12] Jianping Gou, Baosheng Yu, Stephen J. Maybank, and Dacheng Tao. Knowledge distillation: A survey. *International Journal of Computer Vision*, 2021.
- [13] Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. MiniLLM: Knowledge distillation of large language models. 2024.
- [14] Etash Guha, Ryan Marten, Sedrick Keh, Negin Raoof, Georgios Smyrnis, Hritik Bansal, Marianna Nezhurina, Jean Mercat, Trung Vu, Zayne Sprague, et al. OpenThoughts: Data recipes for reasoning models. *arXiv preprint arXiv:2506.04178*, 2025.
- [15] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *Nature*, 2025.
- [16] Bingxiang He, Zekai Qu, Zeyuan Liu, Yinghao Chen, Yuxin Zuo, Cheng Qian, Kaiyan Zhang, Weize Chen, Chaojun Xiao, Ganqu Cui, et al. JustRL: Scaling a 1.5B LLM with a simple RL recipe. *arXiv preprint arXiv:2512.16649*, 2025.
- [17] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

- [18] Jonas Hübner, Frederike Lübeck, Lejs Behric, Anton Baumann, Marco Bagatella, Daniel Marta, Ido Hakimi, Idan Shenfeld, Thomas Kleine Buening, Carlos Guestrin, and Andreas Krause. Reinforcement learning via self-distillation. *arXiv preprint arXiv:2601.20802*, 2026.
- [19] Ijun Jang, Jewon Yeom, Juan Yeo, Hyunggu Lim, and Taesup Kim. Stable on-policy distillation through adaptive target reformulation. *arXiv preprint arXiv:2601.07155*, 2026.
- [20] Xiaoqi Jiao, Yichun Yin, Lifeng Shang, Xin Jiang, Xiao Chen, Linlin Li, Fang Wang, and Qun Liu. TinyBERT: Distilling BERT for natural language understanding. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020.
- [21] Woogyoul Jin, Taywon Min, Yongjin Yang, Swanand Ravindra Kadhe, Yi Zhou, Dennis Wei, Nathalie Baracaldo, and Kimin Lee. Entropy-aware on-policy distillation of language models. *arXiv preprint arXiv:2603.07079*, 2026.
- [22] Jeonghye Kim, Xufang Luo, Minbeom Kim, Sangmook Lee, Dohyung Kim, Jiwon Jeon, Dongsheng Li, and Yuqing Yang. Why does self-distillation (sometimes) degrade the reasoning capability of LLMs? *arXiv preprint arXiv:2603.24472*, 2026.
- [23] Yoon Kim and Alexander M. Rush. Sequence-level knowledge distillation. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2016.
- [24] Jongwoo Ko, Sara Abdali, Young Jin Kim, Tianyi Chen, and Pashmina Cameron. Scaling reasoning efficiently via relaxed on-policy distillation. *arXiv preprint arXiv:2603.11137*, 2026.
- [25] Jia Li, Edward Beeching, Lewis Tunstall, Ben Lipkin, Roman Soletskyi, Shengyi Huang, Kashif Rasul, Longhui Yu, Albert Q. Jiang, Ziju Shen, Zihan Qin, Bin Dong, Li Zhou, Yann Fleureau, Guillaume Lample, and Stanislas Polu. NuminaMath: The largest public dataset in AI4Maths with 860k pairs of competition math problems and solutions. Hugging Face repository, 2024.
- [26] Yaxuan Li, Yuxin Zuo, Bingxiang He, Jinqian Zhang, Chaojun Xiao, Cheng Qian, Tianyu Yu, Huan-ang Gao, Wenkai Yang, Zhiyuan Liu, and Ning Ding. Rethinking on-policy distillation of large language models: Phenomenology, mechanism, and recipe. *arXiv preprint arXiv:2604.13016*, 2026.
- [27] Yuetai Li, Xiang Yue, Zhangchen Xu, Fengqing Jiang, Luyao Niu, Bill Yuchen Lin, Bhaskar Ramasubramanian, and Radha Poovendran. Small models struggle to learn from strong reasoners. 2025.
- [28] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.
- [29] Kevin Lu and Thinking Machines Lab. On-policy distillation. *Thinking Machines Lab: Connectionism*, 2025. doi: 10.64434/tml.20251026.
- [30] Yun Luo, Zhen Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. An empirical study of catastrophic forgetting in large language models during continual fine-tuning. *IEEE Transactions on Audio, Speech and Language Processing*, 2025.
- [31] Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, et al. WebGPT: Browser-assisted question-answering with human feedback. In *arXiv preprint arXiv:2112.09332*, 2021.
- [32] Emiliano Penalosa, Dheeraj Vattikonda, Nicolas Gontier, Alexandre Lacoste, Laurent Charlin, and Massimo Caccia. Privileged information distillation for language models, 2026.
- [33] Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2011.
- [34] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. In *Proceedings of the 5th Workshop on Energy Efficient Machine Learning and Cognitive Computing (NeurIPS Workshop)*, 2019.

- [35] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- [36] Idan Shenfeld, Jyothish Pari, and Pulkit Agrawal. RL’s razor: Why online reinforcement learning forgets less. *arXiv preprint arXiv:2509.04259*, 2025.
- [37] Idan Shenfeld, Jyothish Pari, and Pulkit Agrawal. RL’s razor: Why online reinforcement learning forgets less. *arXiv preprint arXiv:2509.04259*, 2025.
- [38] Idan Shenfeld, Mehul Damani, Jonas Hübner, and Pulkit Agrawal. Self-distillation enables continual learning. *arXiv preprint arXiv:2601.19897*, 2026.
- [39] Avi Singh, John D. Co-Reyes, Rishabh Agarwal, Ankesh Anand, Piyush Patil, Xavier Garcia, Peter J. Liu, James Harrison, Jaehoon Lee, Kelvin Xu, Aaron Parisi, Abhishek Kumar, Alex Alemi, Alex Rizkowsky, Azade Nova, Ben Adlam, Bernd Bohnet, Gamaleldin Elsayed, Hanie Sedghi, Igor Mordatch, et al. Beyond human data: Scaling self-training for problem-solving with language models. *Transactions on Machine Learning Research*, 2024.
- [40] Charlie Snell, Dan Klein, and Ruiqi Zhong. Learning by distilling context, 2022.
- [41] Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul F. Christiano. Learning to summarize with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [42] Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. *Advances in Neural Information Processing Systems*, 33:5776–5788, 2020.
- [43] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. 2023.
- [44] Xiaomi LLM-Core Team, Bangjun Xiao, Bingquan Xia, Bo Yang, et al. MiMo-V2-Flash technical report. *arXiv preprint arXiv:2601.02780*, 2026.
- [45] Tian Xu, Ziniu Li, and Yang Yu. Error bounds of imitating policies and environments. *Advances in Neural Information Processing Systems*, 33, 2020.
- [46] Wenda Xu, Rujun Han, Zifeng Wang, Long T Le, Dhruv Madeka, Lei Li, William Yang Wang, Rishabh Agarwal, Chen-Yu Lee, and Tomas Pfister. Speculative knowledge distillation: Bridging the teacher-student gap through interleaved sampling. *arXiv preprint arXiv:2410.11325*, 2024.
- [47] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [48] Chenxu Yang, Chuanyu Qin, Qingyi Si, Minghui Chen, Naibin Gu, Dingyu Yao, Zheng Lin, Weiping Wang, Jiaqi Wang, and Nan Duan. Self-distilled RLVR. *arXiv preprint arXiv:2604.03128*, 2026.
- [49] Wenkai Yang, Weijie Liu, Ruobing Xie, Kai Yang, Saiyong Yang, and Yankai Lin. Learning beyond teacher: Generalized on-policy distillation with reward extrapolation. *arXiv preprint arXiv:2602.12125*, 2026.
- [50] Tianzhu Ye, Li Dong, Zewen Chi, Xun Wu, Shaohan Huang, and Furu Wei. Black-box on-policy distillation of large language models, 2026.
- [51] Tianzhu Ye, Li Dong, Xun Wu, Shaohan Huang, and Furu Wei. On-policy context distillation for language models. *arXiv preprint arXiv:2602.12275*, 2026.
- [52] Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, et al. DAPO: An open-source LLM reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.

- [53] Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah Goodman. STaR: Bootstrapping reasoning with reasoning. *Advances in Neural Information Processing Systems*, 2022.
- [54] Chen Zhang, Qiuchi Li, Dawei Song, Zheyu Ye, Yan Gao, and Yao Hu. Towards the law of capacity gap in distilling language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 22504–22528, 2025.
- [55] Siyan Zhao, Zhihui Xie, Mengchen Liu, Jing Huang, Guan Pang, Feiyu Chen, and Aditya Grover. Self-distilled reasoner: On-policy self-distillation for large language models. *arXiv preprint arXiv:2601.18734*, 2026.

This appendix provides additional details to support the reproducibility and interpretation of our results. Appendix A describes the model, data, training, and evaluation configurations used in our experiments. Appendix B presents the prompt templates, ground-truth-guided fallback, prompt perturbation design, and answer extraction protocol. Appendix C includes additional results about early-step BRTS behavior and computational cost. Appendix D discusses the main limitations of BRTS and outlines future directions for extending the method to broader tasks and more diverse teacher-student settings.

A Implementation Details

Model and data configurations Our main experiments pair a JustRL-DeepSeek-1.5B teacher (used as the reward / distillation model) with a same-scale DeepSeek-R1-Distill-Qwen-1.5B student backbone. For the teacher-swap study in Section 4.2, we replace the teacher with DeepSeek-R1-Distill-Qwen-7B while keeping DeepSeek-R1-Distill-Qwen-1.5B as the student. Both teacher and student share the same tokenizer within each pairing, which simplifies the top- K alignment used by the auxiliary branch. Training prompts are drawn from DAPO-Math-17K [52], a math-reasoning corpus with verifiable short-form answers. Each prompt is paired with its ground-truth answer y^* , used by the rollout selector for correctness checking and, when invoked, by the Tier-2 ground-truth-guided re-rollout.

Training Setup. We use AdamW (default betas (0.9, 0.999), weight decay 0.01, gradient clip 1.0) with a constant learning rate of $1e^{-6}$ and no warm-up. The student is trained with token-mean loss aggregation, mini-batch size 64, and PPO micro-batch size 1 per GPU using dynamic batching. KL regularization to a frozen reference is disabled, so the only KL terms in the objective are the student-context distillation loss and the auxiliary teacher-context KL described in Section 3.3. The auxiliary coefficient is fixed at $\lambda = 10$ throughout training. Models are run in `bfloat16`.

The student samples one rollout per prompt with temperature 1.0, repetition penalty 1.0, maximum prompt length 1024 and maximum response length 7168, using vLLM. Teacher rollouts in BRTS are generated with temperature 0.7 and top- $p = 0.95$ (the standard sampling configuration we found to give the highest pass rates for the teacher), with the same maximum response length as the student. Top- $K = 16$ teacher token candidates are extracted at each teacher position for use by the auxiliary branch. For BRTS- N , we draw N unconditioned teacher rollouts per prompt; the optional Tier-2 fallback fires only on prompts where all N unconditioned samples are incorrect, drawing one additional ground-truth-guided teacher rollout for that prompt.

All main training runs use 8 B200 GPUs on a single node (the codebase targets H100/B200-class hardware with `bfloat16` forward and PyTorch SDPA attention). The auxiliary student forward on the selected teacher trajectory adds a modest overhead because that trajectory is typically shorter than a worst-case student rollout.

Evaluation Setup. For evaluation, we sample $k = 4$ solutions per problem with temperature 0.7 and top- $p = 0.95$, with maximum validation response length 31,744 tokens. We evaluate performance on AIME 2024 [25], AIME 2025 [2], and AMC 2023 [25]. For each problem, we sample four solutions and report the average accuracy across samples, the best accuracy obtained when any sampled solution is correct, and the majority-vote accuracy among the sampled solutions. Validation is run every 10 training steps.

B Prompt Templates

This section provides the prompt templates and parsing rules used in our implementation. We include these details for reproducibility. Unless otherwise stated, all templates preserve the original problem statement and final-answer format, and only modify the auxiliary instructions used to sample or validate teacher trajectories.

Tier-2 ground-truth-guided template. When all N Tier-1 unconditioned teacher samples for a prompt are incorrect, BRTS re-prompts the teacher with the ground-truth answer silently provided in context and asks for a natural derivation. The hint string is injected immediately before the assistant turn marker (e.g., `<|Assistant|>` for DeepSeek-R1-distill chats):

Table 5: **Early-step performance under different rollout configurations.** We report mean and majority accuracy at steps 10 and 20 on AMC23 and AIME25. Replacing one student rollout with BRTS improves early performance on AMC23, while adding Tier-2 recovery yields the largest gains on the harder AIME25 benchmark.

$\mathcal{L}_{\text{tea-ctx}}$	Rollout setup			AMC23 mean		AMC23 majority		AIME25 mean		AIME25 majority	
	Tier-2	Student	Teacher / Cand.	Step 10	Step 20	Step 10	Step 20	Step 10	Step 20	Step 10	Step 20
×	×	2	0 / 0	0.6024	0.6386	0.6171	0.6525	0.2330	0.2083	0.2442	0.2228
✓	×	1	1 / 2	0.6175	0.6416	0.6339	0.6663	0.2167	0.2250	0.2323	0.2309
✓	×	1	1 / 4	0.6386	0.6386	0.6520	0.6581	0.2167	0.2083	0.2207	0.2222
✓	✓	1	1 / 2+1	0.6476	0.6446	0.6689	0.6630	0.2583	0.3000	0.2718	0.3117

```
{original prompt}
[SILENT_VALIDATION_KEY -- DO NOT MENTION IN THINKING OR RESPONSE: {gt}]
STRICT RULES about the validation key:
1. NEVER mention, quote, paraphrase, or allude to it anywhere -- not in
   <think>, not in your answer.
2. NEVER say things like ‘the key says’, ‘based on the hint’, ‘the
   answer is given’, ‘I can see the correct answer is’, or any equivalent
   phrasing.
3. Your entire chain of thought must be derived from what you observe in
   the problem.
4. Only use the validation key silently as a final sanity-check after you
   have already reasoned to a conclusion -- never as a starting point or
   shortcut.
```

The resulting Tier-2 rollout is retained only if its extracted answer matches y^* ; samples for which Tier-2 also fails fall through to Tier-3, in which BRTS picks the most overlap-similar Tier-1 rollout as a best-available trace.

Prompt perturbation. For the perturbation study, we modify the surface form of one of the two teacher prompts without changing the problem statement or the requested answer format. Concretely, we append the following instruction to the prompt of one teacher rollout

```
Please reason step by step and rethink in detail before giving the final
answer.
```

This paraphrases or extends the default instruction (e.g., by reordering clauses or adding a reflection cue) so that the two teacher rollouts are mildly decorrelated at the surface level while task semantics, problem statement, and answer format remain unchanged.

Answer extraction. We extract the final answer from the last `\boxed{...}` expression in a rollout. Rollouts that contain no parseable boxed answer are treated as incorrect.

C Additional Results

Early-step behavior. Table 5 compares different rollout configurations at the early training stages. On AMC23, BRTS improves early mean and majority accuracy over the student-only baseline, with the Tier-2 variant achieving the best step-10 performance. The gains are more pronounced on AIME25: when adding Tier-2 raises mean accuracy from 0.2330 to 0.2583 at step 10 and from 0.2083 to 0.3000 at step 20. Majority accuracy shows the same trend. This suggests that, for harder prompts, ground-truth-guided recovery provides more reliable early supervision.

Computational cost. We report median step times using $8 \times \text{B200}$ GPUs, batch size 64, a 1.5B student and JustRL-1.5B teacher, max response length of 7168. The cost of increasing the teacher candidate pool is summarized in Table 6. Since BRTS selects a single auxiliary teacher trajectory, the main additional overhead comes from sampling multiple teacher candidates before selection.

Table 6: Per-step training cost with increasing teacher rollout candidates.

Teacher rollout / Candidates	Step time (s)	Step time (min)
1 / 1	281	4.7
1 / 2	338	5.6
1 / 3	400	6.7
1 / 4	460	7.7

Empirically, increasing the candidate pool from one to two raises the per-step time from 281s to 338s, a moderate overhead of about 59 seconds. The time costs grow approximately linearly, reaching 460s per step when selecting from four candidates. These results highlight a practical trade-off between supervision quality and training efficiency. Importantly, our main results show that even a small candidate pool can improve OPD, indicating that BRTS does not rely on expensive large-scale teacher sampling to be effective.

D Limitations and Future Work.

BRTS studies Best-of- N rollout teacher selection in the context of mathematical reasoning, where final-answer verification provides a clean and controllable testbed for analyzing teacher-rollout selection. A natural next step is to extend this principle to broader reasoning and generation settings where exact ground-truth answers may be unavailable. In such cases, the Tier-2 mechanism need not rely on gold answers directly; it could instead be guided by additional knowledge resources, retrieval-augmented hints, tool feedback, learned verifiers, or stronger models. For example, a larger teacher, such as a 32B model, could provide intermediate guidance for a 1.5B student, helping construct reliable teacher trajectories even when explicit labels are unavailable. This suggests that BRTS can be viewed more generally as a framework for using external guidance to make teacher supervision more reliable and student-compatible.

Another interesting direction is to improve how teacher trajectories are organized and presented to the student. In the current work, BRTS selects teacher rollouts based on correctness first and student alignment second, using a lightweight top- K overlap criterion as the alignment proxy. Future work could explore more expressive measures of teacher-student compatibility. More broadly, this connects to curriculum design in on-policy distillation: instead of training on tasks in a fixed order, the system could automatically identify which problems are easy, learnable, or too difficult for the current student, and adapt teacher supervision accordingly. Ideally, OPD could behave more like a human teacher, first recognizing the student’s current reasoning ability and then conveying harder problems through trajectories that are both reliable and accessible.

Finally, BRTS can be further studied and scaled along multiple dimensions. BRTS can be scaled by increasing the candidate pool used for teacher-trajectory selection and by using more selected trajectories in the student- and teacher-context losses. These extensions may further improve supervision diversity and robustness, while also raising interesting questions about how to balance teacher sampling cost, trajectory quality, and optimization stability. Future work could also investigate adaptive choices of the auxiliary weight, the number of teacher candidates, and the number of trajectories used for distillation, further establishing BRTS as a scalable framework for improving on-policy distillation.